Hui (Hilla) Kuurila-Zhang

AI Safety · Adversarial Machine Learning · AI System Performance Engineering

kay.h.zhang@gmail.com · Oulu, Finland · huizhg.github.io · GitHub · LinkedIn · Google Scholar

SUMMARY

Deep learning researcher (PhD) in computer vision and adversarial machine learning, now working on AI system performance engineering on AMD GPUs.

AI SYSTEM PERFORMANCE ENGINEERING PROJECTS

ESM-2 protein language model on the AMD MI250X · code · write-up *Oct 2026*

- Profiled ESM-2 3B on real Swiss-Prot proteins with PyTorch Profiler and rocprofv3, and raised throughput **2.99×**, to **82% of the speed of light**, with token-budget bucketing and dynamic-shape torch.compile.
- Ranked the remaining fixes with Amdahl's law, then wrote a fused Triton GEMM + bias + GELU kernel that beats torch.compile on the feed-forward layer.

Triton GEMM on the AMD MI250X · code · write-up *2026*

- Wrote a tiled FP16 GEMM in Triton and tuned it from **19 to 107 TFLOPS**, **matching hipBLASLt** on large square shapes, across 18 benchmark shapes that include the layers of the ESM-2 protein language model and a 7B LLM.
- Diagnosed each bottleneck with the roofline, the compiled kernel (registers, LDS, occupancy, MFMA) and rocprofv3.

vLLM inference profiling on the AMD MI250X *2026*

- Deployed and benchmarked Qwen2.5-VL-7B on vLLM (latency and throughput) on the LUMI supercomputer, and profiled it with rocprofv3 to attribute GPU time to attention, prefill and decode kernels.

Technical blog · Inside AMD GPUs: Anatomy of the MI250X · A Triton GEMM on the MI250X · Profiling ESM-2

EXPERIENCE

Doctoral Researcher · University of Oulu *2022–present · Oulu, Finland*

- Trained and optimized CNNs with custom loops and losses, and designed an efficient adversarial training algorithm (BMVC 2023).
- Built an image synthesis pipeline that steers pretrained latent diffusion models at inference time with adaptive, momentum-based guidance to produce photorealistic, semantically controlled adversarial images (IEEE ICIP 2026).
- Benchmarked open-source vision-language models (Qwen3-VL, LLaVA, BLIP-2) on facial expression recognition, served with vLLM, and probed their failure modes under CLIP-based transfer attacks (IEEE Intelligent Systems 2026).
- First author on four papers, one a survey under review. Experiments ran on AMD MI250X (LUMI) and NVIDIA A100.

Master's Thesis Worker · Volvo Cars, ADAS software team *Jan–Oct 2021 · Gothenburg, Sweden*

- Built a physics-based simulation of the environment, the target objects and an ultrasonic sensor array in COMSOL Multiphysics, a simulated world for collecting training data.
- Trained a CNN on the simulated sensor data to classify objects, and wrote optimization algorithms for 3D localization.

SKILLS

GPU performance: Triton · torch.compile · ROCm (rocprofv3, Perfetto) · PyTorch Profiler · roofline, occupancy · vLLM

Machine learning: PyTorch · Hugging Face · CNNs, ViTs, diffusion models, VLMs, LLMs, ESM-2 · adversarial robustness

Programming and tools: Python, C++, Bash · Git, Linux · SLURM, Apptainer, HPC · Claude Code · COMSOL Multiphysics

Languages: Chinese (native) · English (full professional) · Finnish (intermediate) · Swedish (entry)

EDUCATION

PhD, Computer Science · University of Oulu · *Computer vision and AI safety* *2022–2026 (defense early 2027)*

MSc, Computer Science · Umeå University · *Robotics and control* *2019–2021*

BSc, Physics · Umeå University · **BE, Electronics Engineering** · Shenzhen University · *dual degree* *2015–2019*

SELECTED PUBLICATIONS

Wei, H., Yu, H., **Kuurila-Zhang, H.**, & Zhao, G. (2026). How private is private? A comparative study for face de-identification. *NeurIPS*.

Kuurila-Zhang, H., et al. (2026). Beyond the ℓp norm: a survey of unrestricted adversarial examples. *Under review*.

Kuurila-Zhang, H., et al. (2026). Venom: text-driven unrestricted adversarial example generation with diffusion models. *IEEE ICIP*.

Kuurila-Zhang, H., et al. (2026). Evaluating the adversarial robustness of vision–language models for facial expression recognition. *IEEE Intell. Syst.*

Kuurila-Zhang, H., et al. (2023). Adaptive adversarial norm space for efficient adversarial training. *BMVC*.